Exercise Sheet 25

The conventions are those of Lecture 25: $c(u)c(v) + c(v)c(u) = 2q(u,v)$, $q(e_i, f_j) = \tfrac{1}{2}\delta_{ij}$, $S^+ = L(\omega_n)$ and $S^- = L(\omega_{n-1})$ in type D for $n \geq 3$. Exercises 25.1–25.3 build fluency. Exercises 25.4–25.5 reconstruct proof steps, and Exercises 25.6–25.7 transfer the methods.

Exercise 25.1 (The B_2 spinor square). Let $V = \mathbb{C}^5$ and let S be the four-dimensional spin module of B_2, with $\omega_1 = \varepsilon_1$ and $\omega_2 = \tfrac{1}{2}(\varepsilon_1 + \varepsilon_2)$.

(i) List the weights of $S \otimes S$ with multiplicities.

(ii) Prove $S \otimes S \cong \mathbb{C} \oplus V \oplus \Lambda^2 V$, dimensions $16 = 1 + 5 + 10$, by highest weights and the Weyl dimension formula.

(iii) Put $\chi = \operatorname{ch} M$ and define $\psi^2(e^\mu) = e^{2\mu}$. Using

$$\operatorname{ch} \operatorname{Sym}^2 M = \frac{1}{2}(\chi^2 + \psi^2\chi), \qquad \operatorname{ch} \Lambda^2 M = \frac{1}{2}(\chi^2 - \psi^2\chi),$$

decompose $\operatorname{Sym}^2 S$ and $\Lambda^2 S$. What kind of invariant form does S carry?

Exercise 25.2 (D_3 compared with $\mathfrak{sl}_4$). Lecture 24 identified the half-spin modules of $\mathfrak{so}_6$ with $\mathbb{C}^4$ and $(\mathbb{C}^4)^*$ under $\mathfrak{so}_6 \cong \mathfrak{sl}_4$. Compose with $X \mapsto -X^{\mathsf{T}}$ if necessary, so that $S^+ \leftrightarrow \mathbb{C}^4$.

(i) Write the three D_3 chirality rules and their dimensions.

(ii) Compare with $\mathbb{C}^4 \otimes \mathbb{C}^4$ and $\mathbb{C}^4 \otimes (\mathbb{C}^4)^*$, and identify V, $\Lambda^2 V$ and $L(2\omega_3)$ as $\mathfrak{sl}_4$-modules.

(iii) Explain why S^+ is the odd-degree part of ΛW, and why it is not self-dual although V is.

Exercise 25.3 (Clifford matrices in dimension two). Let $n = 1$, so $V = \operatorname{span}(e_1, f_1)$ and $S = \Lambda(\mathbb{C}e_1)$ has ordered basis $(1, e_1)$. Put $u_1 = e_1 + f_1$ and $u_2 = i(e_1 - f_1)$. Write $D(X)$ for the matrix of $\rho_S(X)$ in this basis, with images as columns.

(i) Find the matrices $D(\gamma_1)$, $D(\gamma_2)$ and $D(\gamma_1\gamma_2)$.

(ii) Check that $1, \gamma_1, \gamma_2, \gamma_1\gamma_2$ give a trace-orthogonal basis of $M_2(\mathbb{C})$.

(iii) Find λ with $\Gamma = \lambda\gamma_1\gamma_2$.

(iv) Show $A_{u_1u_2} = -iH_1$ and check $\sigma(A_{u_1u_2}) = -i\sigma(H_1)$ on S.

Exercise 25.4 (The volume element). Let $\Omega = \gamma_1 \cdots \gamma_N$ for an orthonormal basis.

(i) Prove $\Omega^2 = (-1)^{N(N-1)/2}$.

(ii) For $N = 2n + 1$, prove that Ω is central in $\operatorname{Cl}(V)$ and that $\rho_S(\Omega)$ is a scalar μ with $\mu^2 = (-1)^n$. Deduce that $\rho_S(\operatorname{Cl}(V)) = \rho_S(\operatorname{Cl}^0)$.

(iii) For $N = 2n$, deduce $\Gamma = \pm i^n \Omega$. Show that exchanging u_1 and u_2 changes the sign of Ω, and explain the consequence for labelling $\Lambda^n_{\pm} V$.

Exercise 25.5 (D_5 and duality). Let $n = 5$, so $\dim S^{\pm} = 16$ and $V = \mathbb{C}^{10}$.

(i) Write the three chirality rules and check dimensions. Find $\dim L(2\omega_5)$.

(ii) Prove that S^+ has no nonzero invariant bilinear form, while there is a nonzero invariant pairing $S^+ \times S^- \to \mathbb{C}$.

(iii) Clifford multiplication gives an equivariant map $V \to \mathrm{Hom}(S^-, S^+)$. In which tensor square does this copy of V appear?

Exercise 25.6 (Triality calculations). Work in D_4 with the notation of Lecture 25.

(i) Choose π with $\pi(4) = 1$, $\pi(3) = 4$, $\pi(1) = 3$. Deduce $8_v \otimes 8_s \cong 8_c \oplus L(\omega_1 + \omega_4)$ from $8_s \otimes 8_c \cong 8_v \oplus 56_v$.

(ii) Compute $\dim L(\omega_1 + \omega_4)$ by the Weyl dimension formula.

(iii) Show that $m(v \otimes \psi) = \rho_S(c(v))\psi$ is a surjective equivariant map $8_v \otimes 8_s \to 8_c$.

Exercise 25.7 (A character proof for B_n). Let $N = 2n + 1$ and $x_i = e^{\varepsilon_i}$.

(i) Show $\mathrm{ch}(S \otimes S) = \prod_{i=1}^{n} (x_i + 2 + x_i^{-1})$.

(ii) Show $\sum_k t^k \,\mathrm{ch}\, \Lambda^k V = (1+t) \prod_{i=1}^{n} (1 + t x_i)(1 + t x_i^{-1})$.

(iii) Deduce $S \otimes S \cong \bigoplus_{k=0}^{n} \Lambda^k V$ without the Clifford decomposition of $\mathrm{End}(S)$, using $\Lambda^k V \cong \Lambda^{N-k} V$.